Chapter 14 exercises

DM14-E01

Find all units and nonzero zero divisors in $\mathbb{Z}_8$, and compute the inverse of 3.

DM14-E02

Fix an integer a. Determine whether polynomial evaluation $ev_a : \mathbb{Z}[x] \to \mathbb{Z}_{12}, p(x) \mapsto [p(a)]_{12}$, is a ring homomorphism. Find its entire kernel and image. What changes if the codomain is $\mathbb{Z}$? Give a quotient-ring isomorphism from the first isomorphism theorem in each case.

DM14-E03

Use the extended Euclidean algorithm to solve $17x \equiv 5 \pmod{43}$.

DM14-E04

Solve $x \equiv 1 \pmod{4}, x \equiv 2 \pmod{5}, x \equiv 3 \pmod{7}$, and find the least nonnegative solution.

DM14-E05

Let $I = \{2k : k \in \mathbb{Z}\}$ and $J = \{[0], [2], [4]\}$. Decide whether they are ideals in their respective rings $\mathbb{Z}$ and $\mathbb{Z}_6$. Determine the structure of $\mathbb{Z}_6/J$.

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DM14-E06

Prove that if R is an integral domain and $a, b \neq 0$, then $ab \neq 0$. Give a counterexample in a ring that is not an integral domain.

DM14-E07

Solve $10x \equiv 14 \pmod{26}$ completely, listing all least nonnegative solutions. Then solve $x \equiv 3 \pmod{10}, x \equiv 8 \pmod{15}$. If solutions exist, identify the correct modulus for uniqueness; otherwise disprove compatibility. Finally replace the second residue 8 by 9 and decide again.

DM14-E08

Find all general ring homomorphisms $\varphi : \mathbb{Z}_4 \to \mathbb{Z}_6$, without assuming preservation of 1. First prove that $e = \varphi([1]_4)$ determines the map, and rigorously derive the necessary conditions on e. Find every kernel and image and identify any unital maps. Testing a single formula is not a complete classification.

DM14-E09

In $M_2(\mathbb{Z})$, let $S = \{\mathrm{diag}(a, 0) : a \in \mathbb{Z}\}$. Prove that it is a subring, find its own multiplicative identity, and decide whether it is a unital subring or ideal of the ambient ring. Find the inverses of

$$P = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}, \qquad Q = \begin{pmatrix} 1 & 2 \\ 2 & 6 \end{pmatrix}$$

in $M_2(\mathbb{Q})$. Which are also units of $M_2(\mathbb{Z})$? Explain the integer restriction on the determinant.

DM14-E10

On $\mathbb{Q}$, define $a \oplus b = a + b - 5$ and $a \otimes b = ab - 5a - 5b + 30$. Instead of expanding every axiom, construct an explicit ring isomorphism to ordinary $\mathbb{Q}$. Find the new zero, multiplicative identity, additive inverse of arbitrary a, invertible elements and their inverses. Compute and check the inverse of 7.

DM14-E11

Construct an isomorphism $\mathbb{Z}_{18} \cong \mathbb{Z}_2 \times \mathbb{Z}_9$ and its inverse. Find every idempotent $x^2 = x$, prove the list complete, and identify the units and nonzero zero divisors among them. Do not merely test all 18 values. Also count and list all units, relating the result to the gcd criterion.

References and attribution

Lin Cing, handwritten discrete mathematics notes. Ralph P. Grimaldi, Discrete and Combinatorial Mathematics, 5th ed., Ch. 14.

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